

L-parameters

Kabiz Kapoor

So. Brief Refresher: Dual groups

G conn'd reductive group over $E (= \mathbb{A}_p)$

In case G is split: E split max. $T \subset G$

Up to isom, (G, T) is determined by the following data:

RD: Root datum $(R, X, R^\vee, X^\vee)$

- $X, X^\vee$ free abelian groups of finite rank, w

$$\langle \cdot, \cdot \rangle: X \times X^\vee \rightarrow \mathbb{Z}$$

- $R \overset{\text{finite}}{\subset} X$ roots

- $R^\vee \overset{\text{finite}}{\subset} X^\vee$ (roots)

$$(\cdot)^\vee: R \xrightarrow{\sim} R^\vee$$

$$\alpha \mapsto \alpha^\vee$$

A1: $\langle \alpha, \alpha^\vee \rangle = 2$

A2: For any $S_\alpha: X \rightarrow X$

$$x \mapsto x - \langle x, \alpha^\vee \rangle \alpha$$

$$S_{\alpha^\vee}: X^\vee \rightarrow X^\vee$$

$$y \mapsto y - \langle y, \alpha \rangle \alpha^\vee$$

we need:

$$S_\alpha(R) = R, \quad S_{\alpha^\vee}(R^\vee) = R^\vee$$

Isomorphism: $\phi: X_1 \xrightarrow{\sim} X_2, \quad \phi^\vee: R_2^\vee \xrightarrow{\sim} R_1^\vee$

$$R_1 \xrightarrow{\sim} X_2$$

$\{ \text{Iso classes of } (G, T) \}$

$\downarrow$

$\{ \text{Iso classes of RD} \}$

$$(G, T): (R, X, R^\vee, X^\vee) \rightarrow (R^\vee, X^\vee, R, X) \rightarrow (\hat{G}, \hat{T})$$

Point: For general G , (pinned) root datum of $G_{\overline{\mathbb{Q}_\ell}}$ will inherit an action

of $\text{Gal}_{\overline{\mathbb{Q}_\ell}} = \Gamma_E$. This will define a cts action $\Gamma_E = \text{Gal}_E \rightarrow \text{Aut}(\hat{G}(\overline{\mathbb{Q}_\ell}))$

which is trivial if G is split,
always factors through a finite quotient

eg. $\hat{GL}_n = GL_n, \quad \hat{SL}_n = PGL_n, \quad \hat{PGL}_n = SL_n, \dots$



§ 1. L-parameters

W_E Weil group discrete topology

$$\begin{array}{ccccccc} 1 & \rightarrow & I_E & \rightarrow & W_E & \rightarrow & \mathbb{Z} \rightarrow 1 \\ & & \parallel & & \downarrow & \lrcorner & \downarrow \\ 1 & \rightarrow & I_E & \rightarrow & \Gamma_E & \rightarrow & \hat{\mathbb{Z}} \rightarrow 1 \end{array}$$

$$I_E \overset{\text{open}}{\subset} W_E, \quad W_E \approx \coprod_{\mathbb{Z}} I_E$$

Def. An L-parameter is a cts group homomorphism

$${}^L\varphi: W_E \rightarrow \hat{G}(\overline{\mathbb{Q}_\ell}) \rtimes W_E \quad \text{s.t.} \quad \pi_{W_E} {}^L\varphi = \text{id}.$$

$$L\varphi(v) = (\varphi(v), v)$$

$$\varphi(vw) = \varphi(v) \cdot {}^v\varphi(w)$$

$$L\text{-parameter} \longleftrightarrow Z_{\text{cts}}^1(W_E; \hat{G}(\bar{G}_E)) \quad , \quad \ell \neq p$$

$$Z_{\text{cts}}^1(W_E; \hat{G}) : \mathbb{Z}_\ell\text{-alg} \rightarrow \text{Set}$$

$$\Delta \longmapsto Z_{\text{cts}}^1(W_E; \hat{G}(\Delta))$$

$\uparrow$

Discrete topology?

ℓ -adic topology?

Idea: Any f.g. $\mathbb{Z}_\ell$ -submodule $M \neq 0$ of $\mathbb{Q}_\ell : M \subset \ell^{-N}\mathbb{Z}_\ell$

$$\mathbb{Z}_\ell \text{ PID} \longrightarrow M = \ell^k \mathbb{Z}_\ell \quad , \quad k \in \mathbb{Z}$$

$$\mathbb{Q}_\ell = \varinjlim_{n \rightarrow \infty} \ell^{-n} \mathbb{Z}_\ell$$

U open in filtered colimit topology iff $U \cap \ell^{-n} \mathbb{Z}_\ell$ open

in $\ell^{-n} \mathbb{Z}_\ell$, $\forall n$

open balls in metric topology.

Δ module over $\mathbb{Z}_\ell$.

$$\Delta = \text{fil. colim } \Delta'$$

f.g. $\mathbb{Z}_\ell$ -submod Δ'

$$\Delta' \subset \Delta$$

ℓ -adic topology

$$Z_{\text{cts}}^1(W_E; \hat{G}) : \mathbb{Z}_\ell\text{-alg} \rightarrow \text{Set}$$

moduli problem

Q How do we remove "cts"?

Step 1: control the wild inertia

Step 2: pass to a discrete "skeleton".

Step 1.

$$1 \rightarrow I_E \rightarrow W_E \rightarrow \mathbb{Z} \rightarrow 1$$

$P_E \subset I_E$ subgroup of wild inertia:

pro- p Sylow subgp of I_E

$$1 \rightarrow \frac{I_E}{P_E} = \prod_{\ell \neq p} \mathbb{Z}_{\ell}(1) \rightarrow \frac{W_E}{P_E} \rightarrow \mathbb{Z} \rightarrow 1$$

$W_E \cong \hat{G}$ factored through a finite quotient

action vanishes on an open subgp V of finite index

$P_E \cap V$ (also pro- p)

$\cap_n P_E \cap V \leftarrow \text{pro-}p$

$\varphi: W_E \rightarrow \hat{G}(\Lambda)$ (cts) homomorphism

$\varphi|_{P_E \cap V}$ open kernel: $\Lambda = \mathbb{Q}_\ell$, $\hat{G}(\mathbb{Q}_\ell) \subset G_n(\mathbb{Q}_\ell)$

$\varphi|_{P_E \cap V}$ on $\varphi^{-1}(K)$ vanishes.

K open pro- ℓ of $G_n(\mathbb{Q}_\ell)$

$Z_{cts}^1(W_E; \hat{G}) = \bigcup_{\substack{p \text{ open} \\ \text{subgp} \\ \text{of } P_E}} Z_{cts}^1(W_E/p; \hat{G})$ transition maps are both open and closed immersions

$$Z_{cts}^1(W_E, \hat{G}) = \coprod_{\substack{\text{some } p \\ \text{open subgroups} \\ P_E}} Z_{cts}^1(W/P; \hat{G}) \quad (?)$$

$$Z_{cts}^1(W_E/P; \hat{G})$$

Step 2. Pass to a discrete "skeleton" of W_E/P .

σ any lift of Frobenius to W_E .

τ any lift of a generator for tame inertia

$W \subset W_E/P$ generated by $P' = P_E/P, \tau, \sigma$.

$$0 \rightarrow I \rightarrow W \rightarrow \sigma^{\mathbb{Z}} \rightarrow 0$$

I sits inside

$$0 \rightarrow P' \rightarrow I \rightarrow \tau^{\mathbb{Z}[\frac{1}{p}]} \rightarrow 0$$

$\uparrow$
 finite p -group

In W/P' elts σ, τ will satisfy $\sigma^{-1}\tau\sigma = \tau^q, \quad q = |K_E|$.

$$\chi = \sigma\tau\sigma^{-1}$$

$$\sigma^{-1}\chi\sigma = \tau$$

$$\sigma^{-1}\chi^q\sigma = \tau^q = \sigma^{-1}\tau\sigma$$

$$\chi^q = \tau$$

Using that $W \subset^{\text{dense}} W_E/P$, any cts cocycle in

$$Z_{\text{cts}}^1(W_E/P; \hat{G}(\Lambda)) \quad \text{det. by value on } W.$$

Converse: Every 1-cocycle $W \rightarrow \hat{G}(\Lambda)$ extends uniquely to a

$$\text{(cts 1-cocycle)} \quad \frac{W_E}{P} \rightarrow \hat{G}(\Lambda)$$

$$Z_{\text{cts}}^1(W_E/P; \hat{G}) = Z^1(W; \hat{G})$$

$\uparrow$
(twisted) representation schemes

$$\text{Spa} \quad \frac{\mathbb{Z}_\ell[(x_{ij})_1, \dots, (x_{ij})_g]}{\text{rel'n and impose } \hat{G} \text{ cocycles}}$$

$\uparrow$

flat and rel. local complete intersection of $\dim \dim \hat{G}$ over $\mathbb{Z}_\ell$.

provided II same holds for $Z_{\text{cts}}^1(W_E, \hat{G})$.

$\curvearrowright$

§ 3. Comments on coarse moduli:

$$Z_{\text{cts}}^1(W_E; \hat{G}) // \hat{G}$$

$$\phi \in Z_{\text{cts}}^1(W_E, \hat{G})$$

$$g \in \hat{G} \text{ acts via } (g \circ \phi)(w) = g \cdot \phi(w) \cdot ({}^w g)^{-1}$$

On every con'd comp., $\text{Spec } A \subset \mathbb{Z}_{\text{cts}}^1(W_E, \hat{G})$

$$\text{Spec } A^{\hat{G}} \subset \mathbb{Z}_{\text{cts}}^1(W_E; \hat{G}) // \hat{G}$$

$\int$
GIT

$L > \mathbb{Z}_\ell$ sep. closed field.

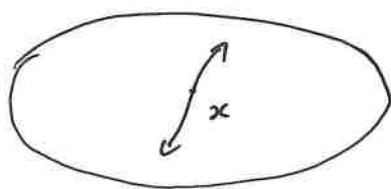
$$\left(\mathbb{Z}_{\text{cts}}^1(W_E; \hat{G}) // \hat{G} \right)(L) \xrightarrow{\sim} \text{closed } \hat{G}(L)\text{-orbits of } \varphi \in \mathbb{Z}_{\text{cts}}^1$$

φ has closed $\hat{G}(L)$ -orbit iff it is semisimple.

Def'n SS:
 $[\varphi] \in \mathbb{Z}_{\text{cts}}^1(W_E)$ contained in $P \rtimes W_E \Rightarrow \varphi(W_E)$ contained in some

$\downarrow$
 parabolic
 for $\hat{G}(L)$

$M \rtimes W_E$
 $\downarrow$
 Levi



$\supset \hat{G}(L)$

(Hilbert - Mumford
criterion)

$$\lambda(t): G_m \rightarrow \hat{G}$$

$\hat{G}(L) \cdot x$ closed iff $\lim_{t \rightarrow 0} \lambda(t) \cdot x$ exists & lies in $\hat{G} \cdot x$.

$G = GL_2$ acting on $\text{Mat}_{2 \times 2}(\mathbb{Z})$

$$\lambda(t) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

$$\lim_{t \rightarrow 0} \lambda(t) M \lambda(t)^{-1} = \lim_{t \rightarrow 0} \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$\mathcal{O}(\mathbb{Z}^1(W; \hat{G}))^{\hat{G}}$: Excursion operators (invert ℓ , or $\ell \neq \pi(\hat{G})_{\text{tors}}$)

$\hat{G} = GL_n$, $W = \text{f.g. group}$.

free gp $\rightarrow F_g \rightarrow W$
on g letters

$$\mathcal{O}(\mathbb{Z}^1(F_g; \hat{G}))^{\hat{G}} \longrightarrow \mathcal{O}(\mathbb{Z}^1(W, \hat{G}))^{\hat{G}} \quad (\text{assump on } \ell)$$

Schur-Weyl duality $\hat{G} = GL_n$

Invariants gen. by traces in the words of F_g .

Excursion operators: $I = \text{finite set}$ $(r_i)_{i \in I} \in W^I$, V rep'n of $(\hat{G} \times W)^I$

$$\alpha: 1 \rightarrow V|_{\hat{G}}$$

$$\beta: V|_{\hat{G}} \rightarrow 1$$

$$\langle \beta, (\phi(r_1), \dots, \phi(r_n)), \alpha \rangle$$

$$I = \{1, 2\}, \quad \hat{G} = GL_n(\Lambda)$$

$$V \simeq \text{Std} \otimes \text{Std}^V \simeq \text{End}(\Lambda^n), \quad (A \cdot B) \cdot M = A M B^{-1}$$

$$\alpha = 1, \quad \beta \in V^* \text{ invt} \quad Y_1 = r \in W, \quad Y_2 = 1 \in W$$

$\searrow$ trace.

$$S(\dots) = \left[\phi \mapsto \text{Tr}(\phi(r)) \right].$$